

Section 7.2

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25010006

T03

1. a) $\ln 0.75 = \ln \frac{3}{4} = \ln 3 - 2 \ln 2$ d) $\ln \sqrt[3]{9} = \ln 3^{\frac{2}{3}} = \frac{2}{3} \ln 3$
- b) $\ln \frac{9}{4} = \ln \frac{2^2}{3^2} = 2 \ln 2 - 2 \ln 3$ e) $\ln 3\sqrt{2} = \ln 3 \cdot 2^{\frac{1}{2}} = \ln 3 + \frac{1}{2} \ln 2$
- c) $\ln \frac{1}{2} = -\ln 2$ f) $\ln \sqrt{3.5} = \ln \sqrt{\frac{27}{2}} = \ln 3^{\frac{3}{2}} \cdot 2^{\frac{1}{2}} = 3 \ln 3 + \ln 2$

4. a) $\ln \sec \theta + \ln \cos \theta = \ln(\sec \theta \cdot \cos \theta) = \ln 1 = 0$
- b) $\ln(8x+4) - 2 \ln 2 = \ln(8x+4) - \ln 4 = \ln\left(\frac{8x+4}{4}\right) = \ln(2x+1)$
- c) $3 \ln \sqrt[3]{t^2-1} - \ln(t+1) = \ln(t^2-1) - \ln(t+1) = \ln\left(\frac{t^2-1}{t+1}\right) = \ln(t-1)$

$$5. \frac{dy}{dx} = \frac{d}{dx}(\ln x + \ln 3) = \frac{1}{x} + 0 = \frac{1}{x}$$

$$8. \frac{dy}{dt} = \frac{d}{dt}\left(\frac{3}{2} \ln t\right) = \frac{3}{2t}$$

$$9. \frac{dy}{dx} = \frac{d}{dx}(\ln 3 - \ln x) = -\frac{1}{x}$$

$$21. \frac{dy}{dx} = \frac{d}{dx}\left(1 - \frac{1}{1+\ln x}\right) = 0 + \frac{1}{(1+\ln x)^2} \cdot \frac{1}{x} = \frac{1}{x(1+\ln x)^2}$$

$$23. \frac{dy}{dx} = \frac{1}{\ln x} \cdot \frac{1}{x} = \frac{1}{x \ln x}$$

$$34. \frac{dy}{dx} = \frac{d}{dx}\left[\frac{5}{2} \ln(x+1) - 10 \ln(x+2)\right] = \frac{5}{2(x+1)} - \frac{10}{x+2}$$

$$39. u = y^2 - 25 \rightarrow du = 2y dy \rightarrow \int \frac{2y dy}{y^2 - 25} = \int \frac{1}{u} du = \ln u + C = \ln(y^2 - 25) + C$$

$$42. u = 1 - 4 \cos \theta \rightarrow du = 4 \sin \theta d\theta \rightarrow \int_0^{\pi/3} \frac{4 \sin \theta}{1 - 4 \cos \theta} d\theta = -\int_{-3}^{-1} \frac{1}{u} du = [\ln |u|]_{-3}^{-1} = -\ln 3$$

$$\theta = 0 \rightarrow u = 1 - 4(1) = -3, \theta = \frac{\pi}{3} \rightarrow u = 1 - 4\left(\frac{1}{2}\right) = -1$$

$$45. u = \ln x \rightarrow du = \frac{1}{x} dx \rightarrow \int \frac{dx}{x(\ln x)^2} = \frac{\ln^4}{\ln^2} \int \frac{1}{u^2} du = \left[-\frac{1}{u}\right]_{\ln^2}^{\ln^4} = \frac{1}{\ln 2} - \frac{1}{\ln 4}$$

$$53. u = 1 + \sqrt{x} \rightarrow du = \frac{1}{2\sqrt{x}} dx \therefore \int \frac{dx}{2\sqrt{x} + 2x} = \int \frac{1}{2\sqrt{x}} \cdot \frac{1}{2\sqrt{x}} dx = \int \frac{1}{u} du = \ln u + C = \ln(1 + \sqrt{x}) + C$$

$$56. \ln y = \frac{1}{2} \ln(x^2+1) + \ln(x-1)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2(x^2+1)} \cdot 2x + \frac{1}{x-1} \quad \left(\frac{d}{dx} \text{ on both sides}\right)$$

$$\frac{dy}{dx} = \sqrt{(x^2+1)(x-1)^2} \cdot \left(\frac{x}{x^2+1} + \frac{1}{x-1}\right) = \frac{x(x-1)}{\sqrt{x^2+1}} + \sqrt{x^2+1}$$

$$65. \ln y = \ln x + \frac{1}{2} \ln(x^2+1) - \frac{2}{3} \ln(x+1)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x} + \frac{2x}{2(x^2+1)} - \frac{2}{3(x+1)}$$

$$\frac{dy}{dx} = \frac{x\sqrt{x^2+1}}{(x+1)^{2/3}} \cdot \left(\frac{1}{x} + \frac{x}{x^2+1} - \frac{2}{3(x+1)}\right)$$

$$70. a) f'(x) = 1 - \frac{1}{x} \therefore \text{For } x > 1, \frac{1}{x} < 1 \text{ and } 1 - \frac{1}{x} > 0, \text{ hence } f'(x) > 0 \text{ or } f(x) \text{ is increasing.}$$

$$b) \text{ From } f(1) = 1 - \ln 1 = 1 > 0 \text{ and } f(x) \text{ is increasing for } x > 1, \text{ we get that } f(x) > 0 \text{ for all } x > 1, \text{ or } \ln x < x \text{ for all } x > 1.$$

73.



$$\begin{aligned} \text{Volume} &= \int_0^3 \pi x^2 dy \\ &= \int_0^3 \pi \left(\frac{2}{\sqrt{y+1}}\right)^2 dy \\ &= \pi \int_0^3 \frac{4}{y+1} dy \\ &= 4\pi [\ln(y+1)]_0^3 = 4\pi \ln 4 \end{aligned}$$

$$77. a) \frac{dy}{dx} = \frac{x}{4} - \frac{1}{x} \rightarrow \text{Length} = \int_4^8 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_4^8 \sqrt{1 + \left(\frac{x}{4} - \frac{1}{x}\right)^2} dx = \int_4^8 \sqrt{\frac{x^2}{16} + \frac{1}{2} + \frac{1}{x^2}} dx = \int_4^8 \left(\frac{x}{4} + \frac{1}{x}\right) dx$$

$$= \left[\frac{x^2}{8} + \ln x\right]_4^8 = 6 + \ln 2$$

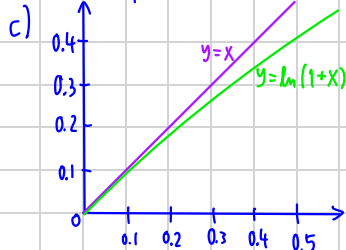
$$b) \frac{dx}{dy} = \frac{y}{8} - \frac{2}{y} \rightarrow \text{Length} = \int_4^{12} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_4^{12} \sqrt{1 + \left(\frac{y}{8} - \frac{2}{y}\right)^2} dy = \int_4^{12} \sqrt{\frac{y^2}{64} + \frac{1}{2} + \frac{4}{y^2}} dy = \int_4^{12} \left(\frac{y}{8} + \frac{2}{y}\right) dy$$

$$= \left[\frac{y^2}{16} + 2 \ln y\right]_4^{12} = 32 + 2 \ln 3$$

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85. a) $L(x) = f(0) + f'(0)(x-0) = \ln(1+0) + \frac{1}{1+0} \cdot (x-0) = 0 + x = x$

b) Error = $|\ln(1+0.1) - 0.1| = 0.00469$



At $x=0$, the approximation of $\ln(1+x)$ seems the best.

$x=0.5$, the approximation of $\ln(1+x)$ seems the least good.

Since the error of approximation is greater as x increases, so the upper bound of error will be at $x=0.5$, whose error equals $|\ln(1+0.5) - 0.5| = 0.09453$

Section 7.3

1. a) $e^{-0.3t} = 27$

$-0.3t = \ln 27 = 3 \ln 3$

$\therefore t = -10 \ln 3$

b) $e^{kt} = \frac{1}{2}$

$kt = \ln \frac{1}{2} = -\ln 2$

$t = -\frac{\ln 2}{k}$

c) $e^{(\ln 0.2)t} = 0.4$

$(\ln 0.2) \cdot t = \ln 0.4 \rightarrow t = \frac{\ln 0.4}{\ln 0.2}$

8. $\frac{dy}{dx} = e^{(4\sqrt{x}+x)} \cdot \left(\frac{2}{\sqrt{x}} + 2x\right)$

14. $y = \ln 3 + \ln \theta - \theta \rightarrow \frac{dy}{d\theta} = \frac{1}{\theta} - 1$

16. $\ln y = 3 \ln \theta - 2\theta + \ln(\cos 5\theta)$

$\frac{1}{y} \frac{dy}{dx} = \frac{3}{\theta} - 2 - \frac{\sin 5\theta}{\cos 5\theta} \cdot 5$

$\frac{dy}{dx} = (\theta^3 e^{-2\theta} \cos 5\theta) \left(\frac{3}{\theta} - 2 - 5 \tan 5\theta\right)$

24. Let $F'(t) = \ln t \therefore y = e^{2x} \int \ln t \, dt = F(e^{2x}) - F(e^{4\sqrt{x}})$

$\therefore \frac{dy}{dx} = F'(e^{2x}) - F'(e^{4\sqrt{x}}) = \ln(e^{2x}) - \ln(e^{4\sqrt{x}}) = 2x - 4\sqrt{x}$

26. $\ln xy = e^{x+y}$

$\frac{1}{x} + \frac{1}{y} \frac{dy}{dx} = e^x \cdot e^y + e^x \cdot (e^y \frac{dy}{dx})$

$\left(\frac{1}{y} - e^{x+y}\right) \frac{dy}{dx} = e^{x+y} - \frac{1}{x} \therefore \frac{dy}{dx} = \frac{e^{x+y} - \frac{1}{x}}{\frac{1}{y} - e^{x+y}}$

27. $e^{2x} = \sin(x+3y)$

$2e^{2x} = \cos(x+3y) \cdot \left(1 + 3 \frac{dy}{dx}\right)$

$1 + 3 \frac{dy}{dx} = \frac{2e^{2x}}{\cos(x+3y)}$

$\therefore \frac{dy}{dx} = \frac{2e^{2x}}{3\cos(x+3y)} - \frac{1}{3}$

29. $\int (e^{3x} + 5e^{-x}) dx = \frac{e^{3x}}{3} - 5e^{-x} + C$

38. $u = -\sqrt{r} \rightarrow du = -\frac{1}{2\sqrt{r}} dr \therefore \int \frac{e^{-\sqrt{r}}}{\sqrt{r}} dr = \int -2e^u du = -2e^u + C = -2e^{-\sqrt{r}} + C$

48. $u = e^x \rightarrow du = e^x \cdot 2x \therefore \int_0^{\ln \pi} 2x e^x \cos(e^x) dx = \int_1^{\pi} \cos u \, du = [\sin u]_1^{\pi} = -\sin 1$

At $x=0 \rightarrow u=e^0=1$, $x=\sqrt{\ln \pi} \rightarrow u=e^{\ln \pi}=\pi$

56. $\frac{dy}{dx} = 3^{-x} (\ln 3)(-1) = -3^{-x} \ln 3$

59. $\frac{dy}{dx} = \pi x^{\pi-1}$

62. $\frac{dy}{dx} = \pi (\ln \theta)^{\pi-1} \cdot \frac{1}{\theta}$

Section 7.3 (Continue)

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$$65. \frac{dy}{dx} = 2^{\sin 3t} \cdot \ln 2 \cdot \cos(3t) \cdot 3 = 3 \ln 2 \cdot 2^{\sin 3t} \cdot \cos(3t)$$

$$74. y = \frac{1}{\ln 5} \left[\frac{\ln 5}{2} (\ln 7 + \ln x - \ln(3x+2)) \right] \therefore \frac{dy}{dx} = \frac{1}{2} \left[0 + \frac{1}{x} - \frac{3}{3x+2} \right]$$

$$75. \frac{dy}{dx} = \sin(\log_7 \theta) + \theta \cos(\log_7 \theta) \cdot \frac{1}{\theta \ln 7} = \sin(\log_7 \theta) + \frac{\cos(\log_7 \theta)}{\ln 7}$$

$$82. y = t \cdot \log_3 (e^{(\sin t)(\ln 3)}) = t \cdot \log_3 (3^{\sin t}) = t \sin t \therefore \frac{dy}{dt} = t \cos t + \sin t$$

$$85. \int 2^{-\theta} d\theta = \left[-\frac{2^{-\theta}}{\ln 2} \right]_0^1 = -\frac{1}{2 \ln 2}$$

$$88. u = \sqrt{x} \rightarrow du = \frac{1}{2\sqrt{x}} dx \therefore \int \frac{2^{\sqrt{x}}}{\sqrt{x}} dx = \int 2 \cdot 2^u du = \left[\frac{2 \cdot 2^u}{\ln 2} \right]_1^2 = \frac{4}{\ln 2}$$

$$91. u = 2x \ln x \rightarrow du = (2 + 2 \ln x) dx, \text{ At } x=2 \rightarrow u=4 \ln 2, \text{ At } x=4 \rightarrow u=8 \ln 4 = 16 \ln 2$$

$$\therefore \int_2^4 e^{2x \ln x} (1 + \ln x) dx = \frac{1}{4 \ln 2} \int_{4 \ln 2}^{16 \ln 2} \frac{1}{2} e^u du = \left[\frac{1}{2} e^u \right]_{4 \ln 2}^{16 \ln 2} = \frac{1}{2} [2^{16} - 2^4] = 32760$$

$$96. \int_1^e x^{\ln 2x - 1} dx = \left[\frac{x^{\ln 2x}}{\ln 2} \right]_1^e = \frac{2-1}{\ln 2} = \frac{1}{\ln 2}$$

$$99. \int_1^4 \frac{\ln 2 \log_2 x}{x} dx = \int_1^4 \frac{\ln x}{x} dx = \int_0^{\ln 4} u du \quad (u = \ln x \rightarrow du = \frac{1}{x} dx) = \left[\frac{u^2}{2} \right]_0^{\ln 4} = \frac{(\ln 4)^2}{2}$$

$$111. \ln y = x \ln(x+1)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{x}{x+1} + \ln(x+1) \rightarrow \frac{dy}{dx} = (x+1)^x \left(\frac{x}{x+1} + \ln(x+1) \right)$$

$$118. \ln y = \ln x \cdot (\ln(\ln x))$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x} (\ln(\ln x)) + \ln x \cdot \left(\frac{1}{\ln x} \cdot \frac{1}{x} \right)$$

$$\therefore \frac{dy}{dx} = (\ln x)^{\ln x} \left(\frac{1}{x} \right) (\ln(\ln x) + 1)$$

$$121. a) \text{ Let } 0 = f'(x) = -xe^{-x} + e^{-x} = e^{-x}(1-x) \therefore x=1$$

Slope: $\leftarrow + \quad - \rightarrow \rightarrow x=1$ as absolute extreme value (maximum) of f

$$b) f'''(x) = -e^{-x} + xe^{-x} - e^{-x} = xe^{-x} - 2e^{-x} = (x-2)e^{-x}$$

$$\therefore f'''(x) = 0 \text{ when } x=2, \text{ and } f'': \leftarrow - \quad + \rightarrow \therefore x=2 \text{ is the inflection point for } f$$

$$123. \text{ Let } 0 = f'(x) = x^2 \cdot \frac{1}{x} + 2x \ln \frac{1}{x} = x(1-2 \ln x)$$

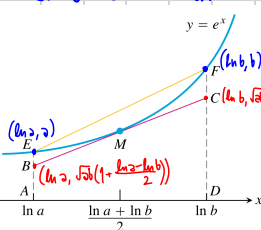
$$\therefore x=0 \text{ or } \ln \frac{1}{x} = -\frac{1}{2} \rightarrow x = \sqrt{e} \therefore f: \leftarrow + \quad - \rightarrow \text{ (f defined on } (0, \infty))$$

$$\text{At } x = \sqrt{e}, f(x) = e \ln \left(\frac{1}{\sqrt{e}} \right) = -\frac{1}{2} e$$

$$131. \frac{dy}{dx} = \frac{1}{\cos x} \cdot (-\sin x) = -\tan x \therefore \text{Length} = \int_0^{\pi/4} \sqrt{1 + (\tan x)^2} dx = \int_0^{\pi/4} \sec x dx = \left[\ln |\sec x + \tan x| \right]_0^{\pi/4} = \ln(\sqrt{2} + 1)$$

$$136. a) \text{ Let } y = e^x \rightarrow \frac{d^2 y}{dx^2} = e^x > 0 \forall x \therefore e^x \text{ is concave up for all values of } x.$$

b)



From the figure, we get

Area under BC < Area under curve < Area under EF

$$\frac{1}{2} (\sqrt{ab} \cdot 2) \cdot (\ln b - \ln a) < \int_{\ln a}^{\ln b} e^x dx < \frac{a+b}{2} (\ln b - \ln a)$$

$$e^{\frac{(\ln a + \ln b)/2} \cdot 2} \cdot (\ln b - \ln a) < \int_{\ln a}^{\ln b} e^x dx < \frac{e^{\ln a} + e^{\ln b}}{2} (\ln b - \ln a)$$

$$c) \text{ From b) we get } \sqrt{ab} (\ln b - \ln a) < [e^x]_{\ln a}^{\ln b} < \frac{a+b}{2} (\ln b - \ln a)$$

$$\text{Divided by } \ln b - \ln a \text{ both sides: } \sqrt{ab} < \frac{b-a}{\ln b - \ln a} < \frac{a+b}{2}$$

Section 7.5

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$$16. \lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} \quad (\text{Type } \frac{0}{0}) = \lim_{x \rightarrow 0} \frac{\cos x - 1}{3x^2} = \lim_{x \rightarrow 0} \frac{-\sin x}{6x} = \lim_{x \rightarrow 0} \frac{-\cos x}{6} = -\frac{1}{6}$$

$$20. \lim_{x \rightarrow 1} \frac{x-1}{\ln x - \sin \pi x} \quad (\text{Type } \frac{0}{0}) = \lim_{x \rightarrow 1} \frac{1}{\frac{1}{x} - \pi \cos \pi x} = \frac{1}{1 - \pi(-1)} = \frac{1}{\pi+1}$$

$$21. \lim_{x \rightarrow 0} \frac{x^2}{\ln(\sec x)} \quad (\text{Type } \frac{0}{0}) = \lim_{x \rightarrow 0} \frac{2x}{\frac{1}{\sec x} \cdot \sec^2 x \tan x} = \lim_{x \rightarrow 0} \frac{2}{\sec^2 x} = 2$$

$$26. \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\pi}{2} - x\right) \tan x \quad (\text{Type } 0 \cdot \infty) = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\left(\frac{\pi}{2} - x\right)}{\cot x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{-1}{-\operatorname{cosec}^2 x} = \frac{1}{\csc^2(\frac{\pi}{2})} = 1$$

$$31. \lim_{x \rightarrow \infty} \frac{\ln(x+1)}{\log_2 x} \quad (\text{Type } \frac{\infty}{\infty}) = \lim_{x \rightarrow \infty} \frac{\frac{1}{x+1}}{\frac{1}{x \ln 2}} = \lim_{x \rightarrow \infty} \frac{x \ln 2}{x+1} = \lim_{x \rightarrow \infty} \frac{\cancel{x} \ln 2}{\cancel{x} + 1} = \ln 2$$

$$38. \lim_{x \rightarrow 0^+} (\ln x - \ln(\sin x)) = \lim_{x \rightarrow 0^+} \ln \left(\frac{x}{\sin x} \right) = \ln \left(\lim_{x \rightarrow 0^+} \frac{x}{\sin x} \right) = \ln \left(\lim_{x \rightarrow 0^+} \frac{1}{\cos x} \right) = \ln 1 = 0$$

$$40. \lim_{x \rightarrow 0^+} \left(\frac{3x+1}{x} - \frac{1}{\sin x} \right) = \lim_{x \rightarrow 0^+} \left(3 + \frac{1}{x} - \frac{1}{\sin x} \right) = 3 + \lim_{x \rightarrow 0^+} \left(\frac{\sin x - x}{x \sin x} \right) = 3 + \lim_{x \rightarrow 0^+} \frac{\cos x - 1}{x \cos x + \sin x}$$

$$= 3 + \lim_{x \rightarrow 0^+} \frac{-\sin x}{-x \sin x + 2 \cos x} = 3 + \frac{0}{2} = 3$$

$$41. \lim_{x \rightarrow 1^+} \left(\frac{1}{x-1} - \frac{1}{\ln x} \right) = \lim_{x \rightarrow 1^+} \left(\frac{\ln x - x + 1}{x \ln x - \ln x} \right) = \lim_{x \rightarrow 1^+} \left(\frac{\frac{1}{x} - 1}{1 + \ln x - \frac{1}{x}} \right) = \lim_{x \rightarrow 1^+} \left(\frac{-\frac{1}{x^2}}{\frac{1}{x} + \frac{1}{x^2}} \right) = \frac{-1}{1+1} = -\frac{1}{2}$$

$$45. \lim_{t \rightarrow \infty} \frac{e^t + t^2}{e^t - t} = \lim_{t \rightarrow \infty} \left(1 + \frac{t^2 + t}{e^t - t} \right) = 1 + \lim_{t \rightarrow \infty} \left(\frac{t^2 + t}{e^t - t} \right) = 1 + \lim_{t \rightarrow \infty} \left(\frac{2t+1}{e^t - 1} \right) = 1 + \lim_{t \rightarrow \infty} \left(\frac{2}{e^t} \right) = 1 + 0 = 1$$

$$51. y = \lim_{x \rightarrow 1^+} x^{\frac{1}{1-x}} \rightarrow \ln y = \lim_{x \rightarrow 1^+} \frac{\ln x}{1-x} = \lim_{x \rightarrow 1^+} \frac{\frac{1}{x}}{-1} = -1 \quad \therefore y = \lim_{x \rightarrow 1^+} x^{\frac{1}{1-x}} = e^{-1} = \frac{1}{e}$$

$$53. y = \lim_{x \rightarrow \infty} (\ln x)^{\frac{1}{x}} \rightarrow \ln y = \lim_{x \rightarrow \infty} \frac{\ln(\ln x)}{x} = \lim_{x \rightarrow \infty} \frac{1}{x \ln x} = 0 \quad \therefore y = \lim_{x \rightarrow \infty} (\ln x)^{\frac{1}{x}} = e^0 = 1$$

$$59. y = \lim_{x \rightarrow 0^+} x^x \rightarrow \ln y = \lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} -x = 0 \quad \therefore y = e^0 = 1$$

$$60. y = \lim_{x \rightarrow 0^+} \left(1 + \frac{1}{x}\right)^x \rightarrow \ln y = \lim_{x \rightarrow 0^+} x \ln \left(1 + \frac{1}{x}\right) = \lim_{x \rightarrow 0^+} \frac{\ln \left(1 + \frac{1}{x}\right)}{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \frac{\frac{x-1}{x+1} \cdot \left(-\frac{1}{x^2}\right)}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} \frac{x}{x+1} = 1 \quad \therefore y = e^1 = e$$

$$61. y = \lim_{x \rightarrow \infty} \left(\frac{x+2}{x-1} \right)^x \rightarrow \ln y = \lim_{x \rightarrow \infty} x \ln \left(1 + \frac{3}{x-1} \right) = \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \frac{3}{x-1} \right)}{\frac{1}{x}} = \lim_{x \rightarrow \infty} \frac{\frac{x-1}{x+2} \cdot \left(-\frac{3}{(x-1)^2} \right)}{-\frac{1}{x^2}} = \lim_{x \rightarrow \infty} \frac{3x^2}{(x-1)(x+2)}$$

$$= \lim_{x \rightarrow \infty} \frac{3}{1 + \frac{1}{x-1} + \frac{2}{x+2}} = 3 \quad \therefore y = \lim_{x \rightarrow \infty} \left(\frac{x+2}{x-1} \right)^x = e^3$$

$$67. \lim_{x \rightarrow \infty} \sqrt{\frac{9x+1}{x+1}} = \sqrt{\lim_{x \rightarrow \infty} \frac{9x+1}{x+1}} = \sqrt{\lim_{x \rightarrow \infty} \frac{9 - \frac{1}{x+1}}{1 + \frac{1}{x+1}}} = \sqrt{9} = 3$$

$$68. \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{\sin x} = \sqrt{\lim_{x \rightarrow 0^+} \frac{x}{\sin^2 x}} = \sqrt{1} = 1$$

75. b is correct. It should be in indeterminate form (ex. $\frac{0}{0}$) to apply the L'Hopital Rule.

$$85. \text{ Let } y = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k \rightarrow \ln y = \lim_{k \rightarrow \infty} k \ln \left(1 + \frac{1}{k}\right) = \lim_{k \rightarrow \infty} \frac{\ln \left(1 + \frac{1}{k}\right)}{\frac{1}{k}} = \lim_{k \rightarrow \infty} \frac{\frac{1}{1+\frac{1}{k}} \cdot \left(-\frac{1}{k^2}\right)}{-\frac{1}{k^2}} = \lim_{k \rightarrow \infty} \frac{r}{1 + \frac{1}{k}} = r$$

$$\therefore y = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k}\right)^k = e^r$$

88. Since $f(0) = 0 = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} e^{-1/x^2}$, so $f(x)$ is continuous at $X=0$.

$$f'(0) = \lim_{r \rightarrow 0} \frac{f(r) - f(0)}{r - 0} = \lim_{r \rightarrow 0} \frac{e^{-1/r^2}}{r} = \lim_{r \rightarrow 0} \frac{\frac{1}{r}}{e^{1/r^2}} = \lim_{r \rightarrow 0} \frac{-\frac{1}{r^2}}{e^{1/r^2} \cdot \left(\frac{2}{r^3}\right)} = \lim_{r \rightarrow 0} \frac{r}{2} \cdot e^{-1/r^2} = 0$$

Section 7.8

1. a) Slower c) Slower e) Slower g) Same rate

b) Slower

d) Faster

f) Slower

h) Slower

9. a) False: $\lim_{x \rightarrow \infty} \frac{x}{x} = 1 \neq 0 \therefore x \neq o(x)$

b) False: $\lim_{x \rightarrow \infty} \frac{x}{x+5} = 1 \neq 0 \therefore x \neq o(x+5)$

c) True: $\frac{x}{x+5} \rightarrow 1$ as $x \rightarrow \infty \therefore x = O(x+5)$

d) True: $\frac{x}{2x} \rightarrow \frac{1}{2}$ as $x \rightarrow \infty \therefore x = O(2x)$

10. a) True: $\frac{1}{\frac{1}{x+3}} = \frac{x}{x+3} \rightarrow 1$ as $x \rightarrow \infty$

b) True: $\frac{\frac{1}{x} + \frac{1}{x^2}}{\frac{1}{x}} = 1 + \frac{1}{x} \rightarrow 1$ as $x \rightarrow \infty$

c) False: $\lim_{x \rightarrow \infty} \frac{\frac{1}{x} - \frac{1}{x^2}}{\frac{1}{x}} = \lim_{x \rightarrow \infty} 1 - \frac{1}{x} = 1 \neq 0$

e) True: $\lim_{x \rightarrow \infty} \frac{e^x}{e^{2x}} = 0 \therefore e^x = o(e^{2x})$

f) True: $\frac{x + \ln x}{x} \rightarrow 1$ as $x \rightarrow \infty \therefore x + \ln x = O(x)$

g) False: $\lim_{x \rightarrow \infty} \frac{\ln x}{\ln 2x} = \lim_{x \rightarrow \infty} \frac{\ln x}{\ln x + \ln 2} = 1 \neq 0$

h) True: $\frac{\sqrt{x+5}}{x} \rightarrow 1$ as $x \rightarrow \infty \therefore \sqrt{x+5} = O(x)$

d) True: $\frac{2 \cos x}{2} \leq \frac{3}{2}$ as x is arbitrarily large

e) True: $\frac{e^x + x}{e^x} \rightarrow 1$ as $x \rightarrow \infty$

f) True: $\lim_{x \rightarrow \infty} \frac{x \ln x}{x^2} = \lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$

g) True: $\lim_{x \rightarrow \infty} \frac{\ln(\ln x)}{\ln x} = \lim_{x \rightarrow \infty} \frac{\frac{1}{\ln x}}{1/x} = \lim_{x \rightarrow \infty} \frac{1}{\ln x} = 0$

h) False: $\lim_{x \rightarrow \infty} \frac{\ln x}{\ln(x^2+1)} = \lim_{x \rightarrow \infty} \frac{\ln x}{2 \ln x + \ln(1+\frac{1}{x^2})} = \frac{1}{2} \neq 0$

11. Since $f(x)$ and $g(x)$ grow at the same rate as $x \rightarrow \infty$, we have $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L$; L const
 $\therefore \frac{f(x)}{g(x)} \rightarrow L$ as $x \rightarrow \infty \therefore f = O(g)$, $\frac{g(x)}{f(x)} \rightarrow \frac{1}{L}$ as $x \rightarrow \infty \therefore g = O(f)$

12. The polynomial $f(x)$ will be of smaller order than polynomial $g(x)$ as $x \rightarrow \infty$ when the degree of $f(x)$ is less than of $g(x)$: let $\text{Deg}(f) = m$, $\text{Deg}(g) = n$; $m \leq n$, $f(x) = x^m \cdot P(x)$, $g(x) = x^n \cdot Q(x)$ and f, g have leading coefficient as a, b respectively $\therefore \lim_{x \rightarrow \infty} P(x) = a$, $\lim_{x \rightarrow \infty} Q(x) = b$

$$\therefore \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{x^m \cdot P(x)}{x^n \cdot Q(x)} = \lim_{x \rightarrow \infty} \frac{1}{x^{n-m}} \cdot \frac{P(x)}{Q(x)} \xrightarrow{a \rightarrow a, b \rightarrow b} = 0$$

$$20. \lim_{x \rightarrow \infty} \frac{e^x}{2n x^n + 2n-1 x^{n-1} + \dots + 2x + 2} \xrightarrow{\text{L'Hopital}} \lim_{x \rightarrow \infty} \frac{e^x}{n 2n x^{n-1} + \dots + 2} \xrightarrow{\text{L'Hopital } n-2 \text{ times}} \lim_{x \rightarrow \infty} \frac{e^x}{n! 2n} = \infty$$

$\therefore$ The function e^x grows faster than any polynomial $2n x^n + \dots + 2x + 2$.

21. a) $\lim_{x \rightarrow \infty} \frac{\ln x}{x^{1/n}} \xrightarrow{\text{L'Hopital}} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{n} x^{\frac{1}{n}-1}} = \lim_{x \rightarrow \infty} \frac{n}{x^{1/n}} = 0 \therefore \ln x$ grows slower than $x^{1/n}$ as $x \rightarrow \infty$ for all $n \in \mathbb{N}$

22. $\lim_{x \rightarrow \infty} \frac{\ln x}{2n x^n + 2n-1 x^{n-1} + \dots + 2x + 2} = \lim_{x \rightarrow \infty} \frac{1}{x(n 2n x^{n-1} + \dots + 2a_2 x + a_1)} = 0 \therefore \ln x$ grows slower than any nonconstant polynomial.